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Reservoir Management and Operations Models: A State-of-the-Art Review

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The objective of this paper is to review the state-of-the-art of mathematical models developed for reservoir operations, including simulation. Algorithms and methods surveyed include linear programming (LP), dynamic programming (DP), nonliner programming (NLP), and simulation. A general overview is first presented. The historical development of each key model is critically reviewed. Conclusions and recommendations for future research are presented.

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INTRODUCTION

During the last 20 years, one of the most important advances made in the field of water resources engineering is the development and adoption of optimization techniques for planning, design, and management of complex water resources systems. The analysis of a complex water resources system may involve thousands of decision variables and constraints. Once the objectives and constraints have been determined, most problems lend themselves to solution techniques developed in the fields of operations research and management sciences. To overcome the problem of dimensionality, various decomposition schemes have been evolved. With the aid of high-speed computers, solutions can be readily obtained to provide decision alternatives which are optimal in some defined sense and which can be used by water resources managers to assist their decision making.

Many successful applications of optimization techniques have been made in reservoir studies, mostly for planning purposes. Discussions during the American Society of Civil Engineers (ASCE) National Workshop in Reservoir Systems Operation, University of Colorado, Boulder, Colorado, August 1979 indicate that a gap may still exist between theories and applications, particularly in the area of real time reservoir operation. Therefore it is extremely important and timely to review and evaluate the state-of-the-art of systems analysis as applied to reservoir operation studies during the last 20 years.

Extensive literature review of the subject of optimization of reservoir operations reveals that no general algorithm exists. The choice of methods depends on the characteristics of the reservoir system being considered, on the availability of data, and on the objectives and constraints specified. In general, the available methods can be classified as follows.

1. Linear programming (LP), including chance-constrained LP, stochastic LP, and stochastic programming with recourse.
2. Dynamic programming (DP), including incremental DP (IDP), discrete differential DP (DDDP), incremental DP and successive approximations (IDPSA), stochastic DP, reliability-constrained DP, differential DP (DDP), and the progressive optimality algorithm.
3. Nonlinear programming.
4. Simulation.

Combinations of the above methods have also been reported in the literature. Critical reviews and in-depth analyses are made of each of the above-mentioned techniques, and the merits and limitations of each technique are assessed. Recommendations on future research needed to advance the state-of-the-art in the field of reservoir management and operation are made.

This review only considers the mathematical models applied to the planning and management of water quantity, specifically, optimal operations. In most cases, the reservoir configuration and capacities are in existence or determined; therefore the very important area of capacity expansion is not included in the review.

GENERAL REQUIREMENTS AND CHARACTERISTICS

The complexities of a multipurpose, multiple-reservoir system generally require release decisions to be determined by an optimization or simulation model. Most optimization models are based on some type of mathematical programming technique. Optimization models have been applied for planning purposes as well as real time operation.

Typical reservoir constraints include conservation (continuity) equations, maximum and minimum storages, maximum and minimum releases, penstock and equipment limitations,

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and contractual, legal, and institutional obligations arising from the various purposes of the system.

The Water Resources Council's Principles and Standard (1973) established two equally important objectives for federal water resources projects: national economic development and environmental quality. In 1979 the objectives were revised to include regional economic development and other social effects. For a reservoir system it is generally accepted that appropriate multiple purposes are water supply for irrigation, industrial and domestic use, hydropower generation, water quality improvement, recreation, fish and wildlife enhancement, flood control, and navigation.

As was pointed out by *Bower et al.* [1962], there should be close correspondence between the performance of the system as simulated (or optimized) at the design stage and that attainable after the system is built. Accordingly, one must use at the design stage an operating procedure (releases) that is consistent with feasible management of the system in being. An optimal operating procedure is needed for the purposes of planning of a complex water resources system. Finding the optimal operating rules of a reservoir system for planning purposes has been a major area of study. The optimal operating rules are used conjunctively with the problem of determining the optimal capacity of a reservoir.

Real time reservoir operation is concerned with the optimal operation of an existing reservoir system, and decisions regarding releases for various purposes have to be made in a considerably shorter time period. A real time optimization model should be easily accessible for updating with new information and have relatively low execution time commensurate with the availability of information and the time interval considered. In addition to the usually explicitly stated categories of system constraints, there is always the formally unstated but very important constraint of continuing operations. That is, whatever the total number of time periods considered for the optimization, the optimal release policy must result in a system state at the end of the last period that is conducive to satisfactory future operations.

The inflow terms in the constraint equations inherently contain hydrologic uncertainty. The uncertainty problem has been treated with the chance-constraint type of formulation or stochastic programming techniques which are useful for planning purposes. For real time operation, streamflow forecasts based on various hydrologic parameters, probability distributions, and measurements are used for uncertain inflow terms. Approaches to solving large-scale mathematical programs applied to reservoir operation have been decomposition (in time and in space), partitioning, aggregation, and various successive approximation techniques.

LINEAR PROGRAMMING (LP) MODELS

LP has been one of the most widely used techniques in water resources management. It is concerned with solving a special type of problem: one in which all relations among the variables are linear, both in constraints and in the objective function to be optimized.

A typical LP model is

$$\min_{X} Z = C^T X \tag{1}$$

subject to

$$AX \geq b \qquad X \geq 0 \tag{2}$$

in which

C n-dimensional vector of objective function coefficients;
X n-dimensional vector of decision variables;
b m-dimensional vector of right-hand sides;
A $m \times n$ matrix of constraint ("technological") coefficients;
T transpose operation.

The application of LP to water resources management vary from relatively simple problems of straightforward allocation of resources to complex situations of operation and management. Under certain assumptions, nonlinear problems can be linearized and solved by iteration or approximation procedures.

If LP is applied to a reservoir operation model, the typical planning objective might be to minimize the capacity (or the cost) of the reservoir that meets all of the constraints or to maximize the total net annual benefit which is represented as a function of storage and release in each presumed operation period. It is required that cost functions be convex and benefit functions concave for the successful application of LP.

Dorfman [1962] demonstrated how LP could be used with three versions of a model, each with increasing complexity. The first version of the model is considered to be the simplest LP application for a simplified river basin planning problem. The most obvious simplifications or limitations of the model are that no overyear storage is considered and that each year is treated like all other years with given long-term average seasonal inflows for two seasons (dry and wet season). The second version of the model uses critical period analysis to estimate the need for overyear storage. In the third version the model treats inflows stochastically. For the above three versions, both storage capacities and target releases were treated as decision variables that must be chosen in order to maximize an economic objective function while satisfying continuity and technological constraints.

Meier and Beightler [1967] introduced a branch-compression technique for decomposing parallel reservoir systems, but their approach did not consider temporal allocation over seasons. *Parikh* [1966] put forth the idea of spatial decomposition by applying DP to subsystems under an initial set of prices. Then, releases over time from all subsystems were allocated over space by using LP, and from which dual values were used to adjust the initial prices assigned to the subsystems. *Roefs and Bodin* [1970] suggested an extension of this methodology to decomposition over time, but substantial computational difficulties were encountered when it was applied to a system of three reservoirs.

Hall and Shephard [1967] developed a DP-LP technique for a river reservoir optimization in which the multiple-reservoir system is decomposed into a master problem and subproblems. The master problem could be seen as a system coordination agency and the subproblems as single reservoir managers. In that work the subproblems were solved by DP. The schedule of releases and energy production were reported to the system coordination agency (master problem) which is LP.

Windsor [1973] developed a methodology employing a recursive LP as the optimization tool for the analysis of a multireservoir flood control system. In the analysis the river basin is divided into a number of routing reaches, and then a set of expressions describing the movement of flood water within the system is formulated. The reduction in magnitude of flood wave as it is transmitted from a river section upstream of a reservoir to a downstream section is related to the method of gate operation and to some extent to the characteristics of the

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river channel. These features are incorporated into the model by using reservoir and channel routing equations. The short-run objective function is to synchronize the reservoir release schedules so as to minimize the total damage cost at all locations being protected, given the updated runoff hydrographs at all significant points in the system.

Becker and Yeh [1974] suggested a combined solution methodology of LP-DP for the determination of optimum real time reservoir operations associated with the California Central Valley Project. Incorporating some of the other objectives into the constraint set, the LP minimizes the loss in potential energy of the stored waters in the reservoirs resulting from any release policy in each period. LP solutions are embedded in a deterministic forward DP for multiple-period optimization.

A combination of LP with stochastic DP was developed by *Takeuchi and Moreau* [1974]. In this model the objective function consists of two parts: immediate economic losses within the month and the expected present value of future losses as a function of end-of-storage levels in the reservoirs. The latter function is estimated by imbedding the linear programming problem in a stochastic DP problem.

Dagli and Miles [1980] proposed a simple solution approach called adaptive planning for a system of four dams in series. In the objective function the annual total head of water retained in the reservoir is maximized. Since the hydraulic head is a function of the volume of water stored in each reservoir, the objective function and some constraints were nonlinear. The problem was linearized, and an LP was solved based on a forecast of future events, with updates at regular intervals to take into account the latest available information.

Stochastic LP Models

The models reviewed so far are predominantly deterministic. Streamflows are taken to be equal to the mean seasonal inflows or from the historical critical period. However, virtually all hydrologic model parameters are uncertain. Deterministic models using selected values of uncertain input can be useful. When important variables are uncertain, a comprehensive analysis needs to evaluate both the expected performance of the system for random events and the magnitude of loss for system failure. Although the uncertainties of some parameters may be taken into account or be alleviated through a sensitivity analysis, however, the procedure does not explicitly consider these uncertainties and may not lead to satisfactory results.

The following reviews several methodologies which can be used for planning purposes when particular parameters, such as inflows, are considered to be nondeterministic.

Stochastic LP for Markov Process

In order to evaluate the value of a flood control reserve storage for hydroelectric or water supply purposes, it seems reasonable to construct a model that takes into account the fact that inflows are subject to random as well as seasonal influences. *Manne* [1962] demonstrated the application of LP for Markov process optimization with a hypothetical single-reservoir example. The basic components of the Markov process are the "state" and the "transition" attained by a "decision." For reservoir operation the inflow and/or the initial storage at the beginning of each time period are the state. The release made in each period is the decision, and the system makes a transition from one state to another in successive time periods. *Thomas and Watermeyer* [1962] extended Manne's work by defining the initial state of the system as both inflow and storage rather than storage alone.

Loucks [1968] developed a stochastic LP model for a single reservoir subject to random, serially correlated, net inflows. The net inflows for each time period were described by a first order Markov chain and transition probabilities of inflows were estimated from historical inflows. The stochastic LP model was applied to a simplified representation of the Finger Lakes within the Oswego River Basin. The objective function used was the minimization of the sum of the expected squared deviation from the desired or "target" reservoir volumes and discharges. All constraints were linear, and chance constraints could also be added to the basic model. *Loucks* [1968] pointed out the dimensionality problem associated with this type of model in real situations which can easily exceed several thousands of constraints.

Three stochastic techniques, namely DP, policy iteration, and LP for the stochastic model incorporating first-order Markov chains have been reviewed and compared by *Loucks and Falkson* [1970] using a simplified numerical example. *Houck and Cohon* [1978] also assumed a discrete Markov structure for the streamflows. They found a design and management policy for a multipurpose multiple-reservoir system by solving two LP problems sequentially in order to approximate a nonlinear formulation.

Stochastic Programming With Recourse

In another version of stochastic optimization, LP methods have been extended to deal with constraints which include random variables. Under this framework it is important to distinguish between the so-called "here and now" and "wait and see" situations. In the latter, the decision maker is allowed to decide on X after having observed the realization of the stochastic events. Therefore for the wait and see situation the optimal value of the objective function may be a random variable (before the occurrence of the random event), and the estimation of its distribution may be a laborious task, but the solution to the problem as to what decision should be taken is straightforward.

In the here and now situation the problem can be formulated as either a stochastic program with recourse or a chanced-constrained stochastic program. In the case of stochastic programming with recourse, the decision is made in at least two stages. A computational procedure was presented for a two-stage LP model by *Dantzig* [1955], in which the activity levels are determined in the first stage, then a corrective action is followed in the second stage. Essentially, the problem considered is that of finding the optimum value of the vector X, in the following program [*Wets*, 1966; *Prekopa*, 1980]:

$$Z = \min \left\{ CX + E\left[\min_{Y} (gY) \right] \right\} \tag{3}$$

subject to

$$AX = b \tag{4}$$

$$TX + WY = p \tag{5}$$

$$X \geq 0 \qquad Y \geq 0 \tag{6}$$

in which $E\{\ \}$ is the expectation with respect to the random vector p', A is a $m \times n$ deterministic matrix; T is a $k \times n$ matrix; W is a $k \times j$ matrix; b, C, and g are $m \times 1$, $1 \times n$, and $1 \times j$ vectors, respectively; X and Y are $n \times 1$ and $j \times 1$ vectors of decision variables; and p is a $k \times 1$ vector. Elements of T, W, and p can be random variables. First, a decision $\bar{X}$ (feasible) is chosen (first stage) to satisfy

$$A\bar{X} = b \qquad X \in K \tag{7}$$

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in which K is the set of such X vectors that there exists at least one Y for which (5) is satisfied for whatever p is realized after the decision for X is made. Each such X will lead to some cost CX, and when the random event p is observed with certainty, the chosen X will finally lead to a recourse action Y (second stage) such that

$$\min_{Y} gY \tag{8}$$

subject to

$$WY = p - T\bar{X} \qquad Y \geq 0 \tag{9}$$

At stage one a feasible solution $\bar{X}$ is found which is also feasible for stage two. At stage two the problem is solved as soon as the random event occurs. The optimum solution of (8) and (9) is a function of p and $\bar{X}$. Let $\phi(\bar{X}, p) = \min_Y \{gY \mid WY = p - T\bar{X}, Y \geq 0\}$, as the solution of (8) and (9), then $E(\phi(\bar{X}, p))$ is interpreted as the arithmetic mean of the optimum solution. If each random variable has a discrete distribution, then we can obtain the first stage and the second-stage optimum solutions simultaneously [*Dantzig and Madansky*, 1961].

In the special case where $W = (I, -I)$, I being the identity matrix, the problem is called stochastic linear programming with single recourse, otherwise it is called stochastic linear programming with fixed recourse. In the literature, stochastic programs with recourse are also referred to as two-stage stochastic programs, indicating that in the first stage a decision on X is taken and in the second stage (after the random event has been observed) a decision on Y is taken according to (8) and (9). Generalization of this concept of increasing the number of stages by observing subsequent stochastic events and making necessary decisions in between leads to multistage stochastic programs.

In a water resource system the second-stage decisions may be represented by loss functions of not meeting supply targets [*Dorfman*, 1962]. *Dupacova* [1980] has applied stochastic programming with recourse to the problem of water resources management and planning.

Stochastic programming with recourse is conceptually straightforward but represents some difficulties in applications. In general, it is easier to solve linear simple recourse problems with random variables being discrete, uniform, normal, or Erlang distributions. Stochastic linear programs with discrete random variables usually lead to deterministic equivalents of large size. Other types of random distributions either require discrete approximations or resolve themselves into nonlinear deterministic equivalents [*Kall*, 1982]. Complete analysis by a stochastic programming with recourse model requires that consequences of recourse actions be modeled and computed for all possible realizations of the random variables. Recourse actions are evaluated by an adequate estimation of losses resulting from random variation, which is difficult, if not impossible, to find. In any event, it is computationally expensive.

Chance-Constrained LP

Another alternative of stochastic programming model which may be attractive to the practitioner is chance-constrained programming, which is a model that reflects the probability conditions on constraints. By this formulation the problem of estimating the loss function may be partially alleviated, even if the establishment of probability levels for constraints is a difficult task. (In fact, this point is still criticized by *Hogan et al.* [1981].) Expressing the chance-constrained formulation in the form of the stochastic programming formulation yields

$$\min_{X} CX \tag{10}$$

subject to

$$AX = b \tag{11}$$

$$Pr\{TX \geq p\} \geq \alpha \tag{12}$$

$$X \geq 0 \tag{13}$$

where $Pr\{\ \}$ denotes probability; α is an $m \times 1$ constant vector with components $\alpha_i (0 \leq \alpha_i \leq 1)$; and C, X, A, b, T, and p are defined as in (3), (4), (5), and (6).

The idea of chance constraints for LP optimization was first put forward by *Charnes et al.* [1958] for determining refinery rates for heating oils to meet stochastic weather dependent demands. The application of chance-constrained LP to reservoir system optimization was initiated by *ReVelle et al.* [1969].

Consider the constraint represented by (12) and consider, for the moment, α a scalar and T deterministic,

$$Pr\{TX \geq p\} \geq \alpha \tag{14}$$

$$0 \leq \alpha \leq 1 \tag{15}$$

If the probability distribution function of the random variable p is known, the above probabilistic constraint can be converted to a deterministic equivalent by using the cumulative probability distribution function of the random variable p, F_p.

Note that

$$Pr\{p \leq TX\} \geq \alpha \tag{16}$$

$$F_p(TX) \geq \alpha \tag{17}$$

The resulting deterministic equivalent is

$$TX \geq F_p^{-1}(\alpha) \tag{18}$$

in which $F_p^{-1}(\alpha)$ is the inverse of the cumulative probability function at the given value of α. If α is chosen to be 0.9, for example, then there will be 0.1 or 10% probability that the constraint as represented by (14) will not be met.

Chance-constrained formulations neither penalize explicitly the constraint violations nor provide recourse action to correct realized constraint violations as a penalty. For this reason, *Hogan et al.* [1981] warn that the practical usefulness of chance-constrained programming as a modeling technique is seriously limited and it should not be regarded as a substitution for stochastic programming with recourse. A counter example has been presented by *Loucks et al.* [1981] which demonstrates that chance-constrained programming does not yield the same solution as stochastic programming with recourse.

To conclude the theories of stochastic programs, it should be emphasized that the main task in decision making under uncertainty is to derive a deterministic equivalent of the stochastic program. If this step is successful, then a standard optimization solution procedure can be used. If a deterministic equivalent cannot be found, Monte Carlo simulation appears to be the most viable approach.

Linear Decision Rules (LDR)

Linear decision rules that relate releases to storage, inflow, and decision parameters are crucial assumptions in the appli-

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cation of chance-constrained LP to reservoir design and operation. *ReVelle et al.* [1969] first proposed the following original LDR for reservoir design and/or operation:

$$R_t = S_{t-1} - b_t \tag{19}$$

where

R_t release during time period t;
S_{t-1} storage at end of time period $t-1$;
b_t decision parameter to be determined.

They found this rule advantageous in implementation in that (1) the release is determined at the beginning of the time period and (2) it eliminates mathematical difficulties in formulating chance constraints.

There are two basic limitations of the rule. First, it yields conservative results, i.e., overly large reservoir capacities. Possible explanations for the conservative results are (1) the rule may not take into account the complete nature of streamflow stochasticity and (2) the LDR is an additional constraint. Second, the solution from an LDR model is not guaranteed to be optimal as it reduces the number of possible operating policies and each flow in each period is considered critical [*Loucks and Dorfman*, 1975].

Loucks [1970] pointed out that the reservoir operation rules discussed by *Young* [1967] were fundamentally different from the original LDR. Young's rules had coefficients for storage and inflow other than 1 and 0, and his conclusions were made in the context of annual releases as opposed to the monthly or seasonal releases considered with the original LDR. Loucks noted that there are many other optimization models for the solution of single reservoir problems with both linear and nonlinear operating policies and questioned the implication that extra reservoir capacity was justified in order to be able to use an LDR derived from a relatively simple model. He commented that the LDR approach yields a simple and easily solved model because of the simplistic nature of the formulation and the simplified objective function, not because the LDR formulation reduces the complexity of the true real world objective function.

Loucks [1970] proposed the following LDR which he termed the "linear release rule":

$$R_t = S_{t-1} + I_t - b_t \tag{20}$$

where I_t is the inflow during time period t. He found that this rule resulted in less conservative results compared to the original LDR. With this rule the release is no longer a promised delivery announced at the beginning of the period but now an adaptive release juggled to match a value fully specified only by the end of the period.

ReVelle and Kirby [1970] modified the original LDR to include evaporation losses using linearized storage area curves and projected storage. They also used the objective of minimizing the probability of violating the minimum flow constraint. *Joeres et al.* [1971] applied the original LDR, chance constrained LP, synthetic streamflow generation, and simulation in modeling the multiple source water supply system for Baltimore. The objective was to minimize the pumping costs of the backup supply. *Nayak and Arora* [1971] applied a modification of the original LDR, which replaced the usual initial storage with a "net initial storage" consisting of initial storage plus upstream reservoir releases scheduled for that period, to a multireservoir system. They also presented partial results of a sensitivity analysis for initial costs, minimum storage levels, freeboard requirements, and minimum and maximum flow requirements.

Eizel [1972] proposed the following LDR which was based on the original idea of *Bryant* [1961]:

$$R_t = (1 - b_t)(I_t + S_{t-1}) + b_t T_t \tag{21}$$

where T_t is the target release for time period t. The resulting chance-constrained model was a nonlinear separable convex programming problem which he solved by the piecewise linear approximation method of separable programming. The limitations of this LDR as cited by *Eisel* [1972] are (1) it presents mathematical complexities which might offset the advantages gained from the generality of the *Bryant* [1961] operations rule; (2) the queuing theory assumptions for estimation of the storage distributions are not strictly observed; and (3) it has convolution problems.

Sneidovich [1980*b*] found that the LDR model proposed by *Eisel* [1972] performed rather poorly and may lead to unduly conservative results. It ignores shortage and overflow, as far as storage levels are concerned, allowing storage levels to take on values outside the feasible domain determined by the capacity. The queuing theory assumptions, though not seriously violated, introduce positive error in the tails of the distribution because the model is based on "theoretical storage level." Since the chance constraints are expressed in terms of the tails, the use of the model may lead toward suboptimal solutions resulting in unduly conservative design and operation policies.

Eastman and ReVelle [1973] studied the effects of the length of decision periods using the original LDR and found that longer decision periods resulted in larger required reservoir capacity. They attributed it to the decrease in speed of response to changes to inflow caused by longer decision intervals.

Lane [1973] used multiple original LDR's conditioned on antecedent rainfall to preserve autocorrelation. He found that the resulting LP's can be extremely large.

LeClerc and Marks [1973] applied the original LDR to a large-scale existing multiple-reservoir system. They used the objectives of minimizing the sum of the drawdowns and minimizing the maximum drawdown. They concluded that the original LDR is physically sound; the solution of the resulting LP, when used in actual operations, causes water to be wasted; and the validity of LDR's is not yet verified.

Loucks and Dorfman [1975] proposed an LDR with a weighting factor λ_t, $0 \le \lambda_t \le 1$ for current inflow. It is a combination of the original LDR and *Loucks* [1970] linear release rule. If $\lambda_t \neq 0$, then the release is no longer a promised delivery announced at the beginning of the period.

ReVelle and Gundelach [1975] proposed the following LDR to incorporate the stochastic nature of inflows:

$$R_t = S_{t-1} + \beta_t I_t - \beta_{t-1} I_{t-1} - \cdots - \beta_{t-k} I_{t-k} - \cdots + b_t \tag{22}$$
$$0 \le \beta_i \le 1 \qquad i = 0, \ldots, t$$

where I_j is the inflow during time period j, $j = 0, \ldots, t$. A priori knowledge of current inflow is required when using this rule and a nonzero value of β_t. By appropriate choice of the parameters they were able to achieve a smaller release variance with this rule but it required slightly larger reservoir capacity when compared to the original LDR. It also presented mathematical complexities. *Gundelach and ReVelle* [1975] derived an algorithm which solves the resulting problem with

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this LDR for the capacity and decision parameters without requiring a computer LP software package.

Houck [1979] used multiple original LDR's conditioned on the previous period's streamflow to relax the feasible region of operating rules allowing improvement in reservoir operation. The method is limited by the amount of data needed to estimate the conditional cummulative distribution functions (CDF's) and the piecewise linearization of the CDF's for inclusion in LP. He found the multi-LDR's resulted in a larger but still manageable model.

Joeres et al. [1981] labeled *Loucks'* [1970] LDR "utopian," since a priori knowledge of I_t at the beginning of time period t would maximize the information content on available supply. They proposed a predictive LDR, similar to Loucks, which replaced I_t with a predicted inflow, I_t, based on a first-order Markov representation of the autocorrelation of streamflow.

Houck and Datta [1981] performed a comparison study to evaluate multiple LDR models. They found multi-LDR models to be less conservative than single LDR models. As the number of LDR's per time period increased, they found that the required capacity decreased. Validation using two simulation models showed the superiority of multi-LDR models over single LDR models.

J. R. Stedinger (unpublished manuscript, 1982) compared the capacities and operating policies resulting from the original LDR-based model, *Loucks* [1970] LDR-based model, and simulation using the standard operating policies (SOP) and the minimum failure frequency policy (MFF). He found that the original LDR performed poorly when estimated capacities were compared to what was actually required during simulation. The capacities required by Loucks' LDR model was found to be more reasonable and roughly equal to those required with the SOP if it is constrained not to allow minimum storage violations and to achieve the target release in each and every season with a 90% reliability. He found that if only an average 90% release reliability is required, then substantially less capacity would be required and furthermore, the MFF policy required even less capacity. When comparing the LDR's operating policy performance with the SOP and MFF policy, Stedinger found that the original LDR policies made very inefficient use of available reservoir capacity. Similar results were obtained when he compared Loucks' LDR operating policy with the SOP and MFF policies. However, he found Loucks' LDR screening models preferable to the original LDR models for the problem he considered.

However, it should be noted that J. R. Stedinger's (unpublished manuscript, 1982) comparison of LDR models with conventional water supply models is somewhat controversial. This is because the models to which the chance-constrained formulations were compared were single purpose, i.e., water supply only, whereas the LDR models are intended for multiple purpose reservoirs with ideal pairing of uses being water supply and flood control (J. L. Cohon, personal communication, 1984).

Models to define optimum operating rules that do not use the linear decision rule have also been reported in the literature. *Colorni and Fronza* [1976], *Simonovic and Marino* [1980], and others have employed reliability programming to find optimum operating rules.

Final Remarks

The University of Tennessee issued a report assessing the status of a class of optimization models designed to aid in the planning and operation of water resource systems [*Rosenthal*, 1980]. Four characteristics of reservoir systems that such models should incorporate were identified: multiple-reservoirs, multiple-time periods, stochastic inflows, and nonseparable benefit functions. One hundred existing mathematical models were cited but none of the models were found to handle effectively all four characteristics. The main reason for this shortcoming was the incompatibility of stochastic inflows and nonseparable benefits. Of these 100 models, there were three chance-constrained-type models [*Joeres et al.*, 1971; *Lane*, 1973; *Askew*, 1975] that could handle the multiple-period and stochastic inflow characteristics effectively. Only the Joeres' et al. chance-constrained LP simulation could handle the multiple reservoirs, and even this was a single reservoir with pumped water backup. None of the three were found capable of handling a nonseparable objective function.

The stochastic nature of water resource systems suggests continued research in the application of stochastic programming methods. The intrinsic capability of chance-constrained programming to incorporate the stochasticity of inflows in a linear program appear to offer some advantage over other stochastic models.

DYNAMIC PROGRAMMING (DP) MODELS

Dynamic programming, a method formulated largely by *Bellman* [1957], is a procedure for optimizing a multistage decision process. DP is used extensively in the optimization of water resource systems [*Buras*, 1966]. The popularity and success of this technique can be attributed to the fact that the nonlinear and stochastic features which characterize a large number of water resources systems can be translated into a DP formulation. In addition, it has the advantage of effectively decomposing highly complex problems with a large number of variables into a series of subproblems which are solved recursively.

It is not unusual to find that a problem can be formulated in more than one way, and part of the art of DP lies in deciding the most efficient formulation for the problem at hand. For example, stages may represent different points in time or in space, and states may be continuous rather than discrete. The key feature of DP application is that it usually is identified as a serial or progressive directed network for an operation or planning problem, respectively [*Hastings*, 1973]. From every state a terminal state is reached in some predetermined number of stages for serial problems, but it is reached in not more than a predetermined number of states for progressive problems.

When the returns are independent and additive, a typical recurrence relation is

$$f_n(x_n) = \max_{d_n} [r_n(x_n, d_n) + f_{n-1}(x_{n-1})] \tag{23}$$

where x is a state variable; d is a decision variable; r is a return function; n is a stage; $x_{n-1} = t_n(x_n, d_n)$ is the stage-to-stage transformation equation; and $f_0(x_0)$ is given for all terminal states.

To decompose a general problem into stages with decisions required at each stage, the value of every stage should satisfy the separability condition and the monotonicity condition [*Nemhauser*, 1966]. The validity of DP regarding separability can be extended by increasing the number of states, but this extension is achieved at the cost of increased computations. For example, reservoir inflows are defined as a state in addition to reservoir storage when the inflows are Markovian rather than random in the transition equation. If the returns

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are additive, multiplicative, or of the minimax or maximin variety, it is sufficient to check that the returns are independent.

Where there is no special reason for choosing either backward or forward formulation, the backward recurrence is normally used. The procedure of making first a backward and then a forward pass is convenient, especially in problems involving time, as it gives the optimal policy in the chronological order; in stochastic problems backward recurrence is essential, since each stage depends on the results of the former stage. However, forward recurrence is advantageous when a deterministic problem has to be solved several times with different planning horizons. This may occur because a plan is periodically reviewed or where the appropriate horizon is unknown and a sensitivity analysis is undertaken. The value can be extended forward in time without repeating previous calculations.

Constraints which restrict only the state or decision space are advantageous in (discrete) DP because they reduce the amount of computation. In contrast, state and decision space constraints can cause considerable procedural difficulties for other optimization techniques. However, when DP is applied to a multiple-reservoir system, the usefulness of the technique is limited by the so-called "curse of dimensionality" which is a strong function of the number of state variables. For computational efficiency, problems should have no more than a few state variables at a time. All methods of dimensionality reduction involve decomposition into subsystems and the use of iterative procedures.

For a reservoir system, studies are typically divided into planning studies and operation studies. Operation studies are further divisible into short-range and long-range operation. For short-range operation (hourly or daily) one may regard both the water demand and the water inflow as deterministic. For the long-range operation (monthly or yearly) the stochastic nature of the inflows might have to be considered. This can be done either by probabilitistic consideration or by a Monte Carlo simulation using synthetically generated equally likely streamflows. DP is extremely well suited by its nature to handle the following types of problems: progressive directed network for capacity expansion problem, deterministic DP for short-range operation, and stochastic DP for long-range operation.

If DP is applied to the determination of reservoir releases, the state variable is the storage and the decision variable is the release. The stage is represented by the time period i. The stage-to-stage transformation is characterized by the continuity equation

$$S_{i+1} = S_i + I_i - R_i - e_i \tag{24}$$

subject to

$$S_{\min} \le S_{i+1} \le S_{\max} \tag{25}$$

and constraints on R.

Suppose an objective function $J(S, R)$ has been chosen for maximization. Note that J in general is a function of release as well as storage. A typical forward DP recursive equation can be written as

$$f_{i+1}(S_{i+1}) = \max_{R_i} [J(R_i, \bar{S}_i) + f_i(S_i)] \tag{26}$$

$$i = 0, 1, 2, \ldots, T$$

where S_0 is the given initial storage, and where

$$\bar{S}_i = (S_{i+1} + S_i)/2$$

The state variable (storage) is generally discretized into a number of feasible states.

Suppose that the inflow sequence is given and the evaporation term e_i is temporarily ignored; the continuity equation now becomes

$$S_{i+1} = S_i + I_i - R_i \tag{27}$$

If S_{i+1} and S_i are chosen, R_i can be directly computed from the above continuity equation. The optimization is over the proper choices of the R_i's. The problem of interpolation is avoided, since the R_i's are computed by fixing the states S_{i+1} and S_i. Solutions are imbedded in the discretized states. The infeasible transitions are discarded in the solution process.

The inclusion of the evaporation term e_i poses no difficulty, since evaporation is a function of the average storage $\bar{S}_i$ which is equal to

$$\tfrac{1}{2}(S_{i+1} + S_i) \tag{28}$$

The recursive equation is carried out until the final stage T is reached. The optimal solutions can then be traced back to determine the consequent releases and storages.

Applications of DP to reservoir optimization have been reported by *Little* [1955], *Hall and Buras* [1971], *Young* [1967], *Meier and Beightler* [1967], *Hall et al.* [1968], *Schweig and Cole* [1968], *Fitch et al.* [1970], *Liu and Tedron* [1973], *Opricovic and Djordjevic* [1976], and *Collins* [1977].

Incremental DP (IDP) and Discrete Differential DP (DDDP)

The use of IDP for reservoir operation studies was reported by *Hall et al.* [1969] and *Trott and Yeh* [1971], and systemized and referred to by *Heidari et al.* [1971] as discrete differential dynamic programming (DDDP). IDP uses the increment concept for the state variables, a concept first introduced by *Larson* [1968]. The major difference between Larson's state increment DP and IDP is the time interval used in the computation, which is variable in the former and fixed in the latter. *Nopmongcol and Askew* [1976] analyzed the difference between IDP and DDDP and concluded that DDDP is the generalization of IDP and the confusion between these terms is most unfortunate.

The IDP procedure starts with an assumed trial state trajectory, which is a sequence of feasible state vectors resulting in a corresponding initial policy, and an initial value of the objective function. The DP recursive equation is then used to examine the neighboring states that are just above and below the trial state trajectory. If any neighboring trajectory is found to give a better value of the objective function, then this new trajectory replaces the trial state trajectory and the procedure continues until convergence takes place. The procedure follows in an iterative path: replacing the initial sequence of states by the new one chosen in each iteration, and then examining the new neighboring states. Convergence to a local optimum is assumed if no new sequence of states can be found to yield a higher value of the objective function.

IDP or DDDP provides a means of alleviating the curse of dimensionality. For example, consider a two-reservoir problem [*Hall et al.*, 1969]. If reservoir 1 takes on 40 feasible states and reservoir 2 takes on 20 feasible states, the DP recursive equation would have to be evaluated at 800 points in each monthly period. Furthermore, in order to carry out the optimization over the decision variables, for which a similar number of combinations exists, it is necessary to compare them for each state. For this two-reservoir problem the

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number of possible state transitions, in advancing from one monthly time period to the next, would be 640,000 (800 × 800). When this problem is solved by IDP, the number of state transitions is reduced to 81. Computer time and storage requirements are also drastically reduced.

For the purpose of obtaining good convergence, *Hall et al.* [1969] suggested two procedures for defining the increments of the state variables. The first was to keep the increments small but constant throughout the iterations; the second was to reduce the increments as the iterations proceeded. In general, strong correlation was found between the number of iterations required for convergence and the size of the increments used for each iteration. The variable increment was found to be more efficient, these increments being reduced as the iteration proceeded. Furthermore, several iterations with a small increment should be allowed at the end of each computation to improve the value of the objective functions.

Turgeon [1982] has demonstrated that IDP may converge to a nonoptimal solution if the same state increment is used for every stage. He then suggested a method to adjust the increment sizes in each stage to obtain the desired results.

Incremental DP with Successive Approximations (IDPSA)

Another way of alleviating the curse of dimensionality is by using Bellman's concept of successive approximations which decomposes an original multiple-state variable DP into a series of subproblems of one state variable in such a manner that the sequence of optimizations over the subproblems converges to the solution of the original problem. *Larson* [1968], *Trott and Yeh* [1971], *Yeh and Trott* [1972], and *Giles and Wunderlick* [1981] have applied this technique to problems involving multiple reservoirs. Yeh and Trott determined the firm water output from the joint operation of a six-reservoir system. The original forward DP had six state variables and six decision variables, and was solved by successive approximations. The advantage of the method of successive approximations is that the solution of the six-dimensional DP problem is obtained by solving a series of one-dimensional DP problems. This provides a considerable reduction in computations, since as the number of state variables of the DP increases, the computations increase linearly instead of exponentially.

The question of convergence of DP with successive approximation has been considered by *Bellman* [1957] and *Bellman and Dreyfus* [1962]. It is easy to show that monotonic convergence is obtained, but convergence to the global optimum cannot be proved and may not occur in general. However, convergence to a true optimum has been proved for three cases of practical interest by *Korsak and Larson* [1970] in discrete time optimal control problems having as many control variables as state variables, where subproblems have a single state variable.

Yakowitz [1983] has demonstrated that if the discretization of the state space is sufficiently fine and if the limiting trajectory is an interior point of the admissible policies, then the state increment DP technique has linear convergence. He has also shown that IDPSA can solve problems which are beyond discrete DP; IDPSA does overcome the curse of dimensionality, and in contrast to differential DP, can be implemented without computation of derivatives. He also implied that there is more room for improvement in computer implementation of IDPSA. Also by starting from different initial solutions the likelihood of finding the true optimum is increased. *Nopmongcol and Askew* [1976] extended the basic successive approximation techniques to a more generalized case of higher-level combinations, such as two- or three-at-a-time combinations.

Stochastic DP

As was mentioned earlier, the inflows into the reservoir can be considered as a Markov process. *Howard* [1960] introduced the concept of returns or rewards, corresponding to the probability transition matrix, into a Markov process. In a multistage Markovian model the objective is generally to maximize the expected return. A typical discretized stochastic DP when applied to a reservoir optimization model has the following form:

$$f_t(S_t, I_{t+1}) = \max_{R_t} \left\{ \sum_{I_t=0}^{I_{t,\max}} P[I_t \mid I_{t+1}][B(R_t) + f_{t-1}, I_t)] \right\} \tag{29}$$

subject to

$$S_{t-1} = S_t + I_t - R_t - e_t \tag{30}$$

$$f_1(S_1, I_2) = \max_{R_1} \left\{ \sum_{I_t=0}^{I_{1,\max}} P[I_1 \mid I_2][B(R_1)] \right\} \tag{31}$$

in which

- $f_t(S_t, I_{t+1})$ expected return from the optimal operation of the system which has t time periods to the end of the planning period;
- S_t storage at the beginning of time period t;
- I_t inflow during time period t;
- B return (or reward) obtained consequent to releasing a quantity of water "R_t" during time period t. B can also be a function of the storage as in the case of hydropower productions;
- $P[I_t \mid I_{t+1}]$ transition probabilities connecting inflow I_t in the tth time period with inflow I_{t+1} in time period $(t + 1)$;
- e_t evaporation loss during time period t;
- t time period index, e.g., months numbered from the end of planning horizon.

The recursive equation uses the fact that inflow during any given time period is related to the preceding one by a conditional probability, $P[I_t \mid I_{t+1}]$, which represents the probability of I_t during the current time period, such that inflow during the preceding time period was I_{t+1}. The right-hand side of the recursive equation states that for a given starting storage level S_t at the beginning of the current time period t and a given inflow I_{t+1} during the preceding time period $t + 1$, one can find a release "R" such that the expected value of the sum of the immediate return and future return is maximized. The expected value is taken over all possible inflow values ($I_t = 0$ to $I_t = I_{t,\max}$) during the current time period t with conditional probability $P[I_t \mid I_{t+1}]$. Since the release during the current time period is a function of S_t and I_{t+1}, it is possible, with given values of S_t and I_{t+1}, to search over all possible values of R and choose the one that maximizes the expected return. A stochastic DP algorithm has to be solved backwards.

Arunkumar and Yeh [1973] used the stochastic DP to maximize firm power output. A penalty function was used for not meeting the specified firm power level. They also proposed a heuristic decomposition approach for a multireservoir system. The approach consists of fixing a stationary policy for $(m - 1)$ reservoirs (say, 2, ..., m) and optimizing with respect to reser-

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voir 1. The optimized policy of reservoir 1 replaces the initial policy of reservoir 1, and reservoir 2 is chosen for optimization while release rules for reservoir 1, 3, ..., m are fixed. The procedure is continued until either the policies do not change or the successive improvement in the infinite time horizon return functions are uniformly bounded by some desired level. The existence of stationary optimal policies has been shown by *Ross* [1970]. Basically, this technique is conceptually similar to the method of successive approximations. While it has not been possible to prove convergence, the method has the advantage over the usual DP approach in that computation and computer storage requirements increase linearly with m as opposed to exponentially. Arunkumar and Yeh applied the decomposition approach to a two-parallel reservoir system, the Shasta and Folsom reservoirs of the California Central Valley Project. The algorithm was started with the determination of optimal release rules for Shasta, independent of Folsom. The "flip-flop" decomposition algorithm was applied until the improvement between successive approximations of the reward function was uniformly bounded by a reasonably small number.

Stochastic DP models also were developed by *Turgeon* [1981*a*, *b*] for the optimization of weekly operating policies of multireservoir hydroelectric power systems. The concept of successive approximations (one-at-a-time and aggregation/decomposition) was also used to alleviate the problem of dimensionality.

Reliability-Constrained DP

In long-range reservoir operation, the trade-off between return and the associated risk is of prime importance. Although both can be considered objectives in reservoir operation, the returns are usually incorporated into the objective function with risk represented in the constraints. In the DP approach these considerations have been formulated as a probabilistic DP with discounting. By probabilistic, we mean that it does not consider the serial dependence of reservoir inflows as in a stochastic DP formulation but considers only independent inflows specified by their probability distribution function. The reliability-constrained or chance-constrained DP formulation has been solved with the penalty function approach or the Lagrangian duality theory of nonlinear programming.

A probabilistic DP model incorporating reliability constraints was first investigated by *Askew* [1974*a*]. He recognized that previous stochastic formulations exercise no direct control over the probability of failure of the reservoir system being modeled. Stochastic DP derived a policy that allowed the system to fail more frequently than the operating authorities were prepared to accept. The probabilistic DP recursive equation is

$$f_n(S_n) = \max_{R_n} \sum_{I_n} P(I_n)\{B(R_n) + rf_{n-1}(S_{n-1})\} \qquad (32)$$

where S_n is the reservoir initial storage for period n; $r = (1 + i)^{-1}$ is the discount factor with a discount rate i; $B(R_n)$ is the net benefit associated with a release R_n, and $P(I_n)$ is the probabiity of (discrete) inflow I_n during period n. A failure is defined as

$$R_n < T_n$$

in which T_n is the target release in period n. Askew used the probabilistic DP with a penalty function, w:

$$f_n(S_n) = \max_{R_n} P(I_n)\{B(R_n) + rf_{n-1}(S_{n-1}) - w\} \qquad (33)$$

in which $w = 0$ for $R_n \geq T_n$, and $w > 0$ for $R_n < T_n$. This has the effect of reducing the expected net benefit associated with releases that give rise to possible shortfalls and tend to cause a more conservative optimum release policy. In that respect, w operates as a chance constraint on the DP model, provided the probability of failure associated with the value of w is computable. Askew accomplished this by simulating the system using a range of w values. He then plotted the expected net benefit as a percentage of the maximum possible ($w = 0$) versus average number of failures.

Klemeš [1975] argued that if objective information on the penalty function could be quantitatively determined, it could have been built into the original benefit function, and the system could have been optimized in a single step. However, for more complex reservoir systems with serial dependence of inflows the analytic technique would be complicated and the two-stage optimization procedure of *Askew* [1974*a*] seemed to be the only feasible option.

Askew [1974*b*] also provided for the concurrent parallel calculation of the true expected net benefit B_n as a function of storage S_n and of the average number of failures F_a, during the system life N, as a function of storage S_n. A systematic trial-and-error method was used to find a value of the penalty function w that yielded a maximum value for $B_n(S_n)$ while obeying the constraints $F_a(S_n) \leq \gamma$, where γ is an integer constant. While the chance constraint is not imbedded in the probabilistic DP formulation, it is satisfied indirectly by an iterative process.

Sneidovich and David [1975] in their comments on *Askew*'s [1947*b*] paper show that by introducing additional system variables, true chance constraints can be formulated:

$$y(n) \leq \gamma \qquad n = 1, 2, \ldots, N \qquad (34)$$

in which $y(n)$, an additional state variable, is the expected number of failures in the life of the system up to the time period n. However, the curse of dimensionality limits the number of such constraints that can be considered for even a single-reservoir system. In a later paper, *Askew* [1975] found that the discount factor r in the chance-constrained DP formulation was a suitable parameter in place of the penalty w for failure used in his earlier work [*Askew*, 1974*a*, *b*]. Variation of this parameter in the iterative optimization procedure is analogous to the imposition of risk premiums when one is investing in less secure ventures: the greater the probability of failure, the greater the premium demanded.

Rossman [1977] demonstrated that the penalty function form of the probabilistic DP approach is limited to problems having expected value type constraints. Using Lagrangian duality theory, dual variables on the reliability constraints were constructed, and the resulting formulation was solved with the probabilistic DP procedure. If there are k eventualities for which there are desired reliability constraints, the probabilistic equation becomes

$$f_n(S_n) = \max_{R_n} \sum_{I_n} P(I_n)\{B(S_n, R_n, I_n)(1 + i)^{1-n} - \sum_{j=1}^{k} \lambda_j \left[G_j(S_n, R_n, I_n) - \frac{\gamma_i}{N} \right] + f_{n-1}(S_{n-1})\} \qquad (35)$$

where λ_j is the Lagrangian multiplier, γ_j is the maximum allowable occurrence for event j over the life of the project N, and $B(S_n, R_n, I_n)$ and $G_j(S_n, R_n, I_n)$ are the net benefits and the reliability constraints, respectively, such that $G_j = 1$ if event j happens, $G_j = 0$ if event j does not happen. Actually, the second term on the right-hand side of the above recursive

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equation is seen to be a penalty function as proposed by *Askew* [1947*a*] but here developed from the Lagrangian formulation.

Reliability constraints can be related to annual events and/or events defined over the entire life of the reservoir. *Sniedovich* [1979] considered a class of reservoir control problems associated with the latter types of reliability constraints. He defined a performance function R as

$$R = \sum_{n=1}^{N} G_n(S_n, R_n, I_n) \tag{36}$$

in which the characteristic function $G_n = 0$ if and only if no failure occurs during the nth period, and $G_n = 1$ if and only if failure does occur during the nth period [*Rossman*, 1977]. Associated with the performance function R, two types of constraints were considered: (1) an expected value type of constraint

$$E[R \mid S_n] \leq \alpha \tag{37}$$

and (2) a probabilistic type of constraint

$$Pr[R \geq m \mid S_n] \leq \beta \tag{38}$$

in which $E[\ ]$ and $Pr[\ ]$ denote expectation and probability, under the release strategy, respectively. He has shown that the probabilistic constraint can always be expressed as an expected value type of constraint by introducing an additional state variable. The optimal solutions resulting from the imposition of the probabilistic reliability constraint yielded a strategy that is extremely conservative prior to the occurrences of the first failure. Recently, the subproblem of mathematical separation of such constraints as variance of the number of failures over the life of a reservoir system was addressed by *Sniedovich* [1980*a*].

Reliability-constrained DP algorithms for reservoir system optimization are well developed for the case of the single-reservoir system for long-range planning purposes. The constraints on the system failure include the types of probability, expected value, and variance related to annual events and/or events defined over the life of the reservoir. However, there are substantial difficulties in developing algorithms that are both truly optimal and computationally tractable for the multiple reservoir system, especially when there exists the interdependency between streamflows into each of the reservoirs. Almost none of the literature is directed to the true stochastic reliability-constrained DP in terms of how severe and/or how long these failures might be. In real time operation, operators of reservoir systems are conservative in that they are wary of implementing optimum release policies based on probabilistic constraints. Their interest is in meeting contract levels which have been established at the planning stage. Thus the present state of reliability-constrained DP limits its application to long-range planning purposes.

Differential DP

The major problems associated with DP when the technique is applied to a multiple-reservoir system are excessive computing time and computer storage requirements. Hence decomposition applied with iterative procedures have been developed to alleviate the dimensionality problem. For example, incremental DP with successive approximation (IDPSA) and discrete differential DP (DDDP) has been used for the operation of a multiple-reservoir system [*Hall et al.*, 1969; *Trott and Yeh*, 1973; *Heidari et al.*, 1971; *Giles and Wunderlick*, 1981]. These methods need a good initial policy (sequence of states) and the convergence of the solution to the global optimum cannot be proved.

For a certain class of DP problems one can take advantage of the special structure of the formulation and avoid the dimensionality problems usually associated with the DP solution. This situation occurs in the case of DP models with linear dynamics and quadratic performance criterion function (usually known as LQP problems). If the objective function (performance criterion) is separable and convex (for minimization problems) and the system can be described solely by a system of linear dynamic equations, then it can easily be shown that the decision is a linear function of the current state, the recurrence relation (23) is a quadratic function of the state variable X_n for a backward solution, and that recursive formulas providing the coefficients of the linear decision can be derived. Therefore given the initial state of the system, an analytical solution can be obtained. The solution methodology can be generalized to handle problems with several state variables without the problem of dimensionality that is associated with the classical discrete DP [*Dreyfus and Law*, 1977].

If the objective function is not quadratic, or the system dynamics are not linear, Taylor series expansion can be used to provide a quadratic approximation for the objective function and linear expressions for the system dynamics around an initial estimate; then an iterative solution procedure can be used for the non-LQP problem. This process constitutes the so-called differential DP [*Jacobson and Mayne*, 1970].

Murray and Yakowitz [1979] proposed a constrained differential DP applicable to a multiple-reservoir system by modifying the differential DP. The constrained differential DP (CDDP) procedure begins with a quadratic approximation of the Nth-stage loss function and finds the optimal solution which minimizes the loss function subject to the linear constraints. For intermediate stages the quadratic approximation is applied to the optimal value function. The quadratic programming problem which arises at every decision stage of an iteration is solved, and those decision coefficients corresponding to active constraints are stored. In implementing the forward run to obtain the successor policy, precautions have to be taken to ensure that no constraints are violated. The successor decision is chosen to be the solution of the quadratic programming problem with a modified objective function to assure an improvement over the original nominal policy.

Gjelsvik [1982] presented a first-order DDP algorithm for computing optimal control for a five-reservoir system, where the stochastic inflow process was approximated by a few discrete disturbance values in each time step. DDP was used to minimize the expected value of the operating cost for the purpose of seasonal planning of water release and power scheduling.

Principle of Progressive Optimality

Turgeon [1981*a*] presented an algorithm based on the principle of progressive optimality for which the state variables do not have to be discretized. The tactical generating problem for a system of hydropower plants located in series on the same river consists in determining the discharges u_i^k from reservoir i in period k where $k = 1, 2, \ldots K$ and $i = 1, 2, \ldots, n$:

$$\min \sum_{k=1}^{K} C(S^k) \tag{39}$$

subject to

$$x_i^k = x_i^{k-1} + q_i^k - u_i^k \qquad x_i^0 = a_i^0 \qquad x_i^K = a_i^K$$

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$$\sum_i H_i(x_i^{k-1}, u_i^k) + S^k = D^k$$
$$0 \le x_i^k \le \bar{x}_i \qquad 0 \le u_i^k \tag{40}$$

where q_i^k is the total inflow to reservoir i in period k, x_i^k is the content of reservoir i (with the capacity $\bar{x}_i$) at the end of period k; $H_i(x_i^{k-1}, u_i^k)$ is the generation of power plant i in period k; S^k is the energy imported with certain cost; D^k is the demand for electrical energy in period k; and $C(S^k)$ is the production cost in period k. The initial and final contents of reservoir i are the values a_i^0 and a_i^K. The value of a_i^K is a function of a_i^0 and of the river flows in the period $[0, K]$ and is obtained by solving the strategic problem.

The progressive optimality algorithm [*Howson and Sancho*, 1975] for the solution of multistate DP problems converges monotonically. The algorithm adjusts $x_1^k, x_2^k, \ldots, x_n^k$ without changing x_i^{k-1} and x_i^{k+1}, $i = 1, \ldots, n$ to minimize $\{C(S^k) + C(X^{k+1})\}$ under the same constraints defined before. Note that the algorithm does not have the recursive equation in terms of the optimal value function and might be considered as a multidimensional continuous version of the IDP procedure. A solution procedure for the constrained nonlinear programming was suggested which can be solved by a direct search method.

Unlike the CDDP procedure, this approach can deal with discontinuous return functions, and the objective function does not have to be linearized or approximated by the quadratic functions. It can also take into account the time of water travel between upstream and downstream reservoirs. *Turgeon* [1981*a*] proposed to use the IDPSA with very coarse grids of points for the state variables to obtain a good trial trajectory and to implement the algorithm in alternate forward and backward fashions.

Final Remarks

In the class of the deterministic DP procedure with separable objective functions for the general allocation problem in discrete time, the subproblem at each stage was solved by linear programming, quadratic programming, or nonlinear programming techniques. The selection of a method for a particular problem depends upon both the nonlinearity associated with the objective function and the complexity associated with constraints. For long-range planning of reservoir systems these approaches need to be extended to a stochastic DP procedure. In real time operation a multisite streamflow forecasting method which is compatible with the chosen DP approach should be incorporated into the overall model building framework.

Decomposition is essential in the application of DP to a "multiple" reservoir system. *Becker and Yeh* [1974] developed an optimization algorithm for the real time monthly operation of a large-scale water resources system. The procedure makes use of the best features of LP and DP in that LP serves for month-to-month optimization (nonlinearities being accounted for by an iteration technique), and DP is used for the selection of an optimal release policy through the specified number of months. The algorithm requires deterministic forecasts or estimates of monthly streamflows, consistent with real time operation. The similarity and difference between the above LP-DP procedure and the constrained differential DP have been discussed by *Yeh* [1982].

Opricovic and Djordjevic [1976] proposed a three-level hierarchical algorithm for the optimization of a single multipurpose reservoir with direct and indirect users: first level, optimize the time distribution of available water; second level, optimize the distribution to direct users in one time interval; third level, allocate water to indirect users. A multiobjective DP was developed to generate nondominated solutions for a reservoir where the number of objectives is small and the number of decisions is large [*Tauxe et al.*, 1979]. Objectives other than the primary objective are treated as constraints, which necessitates the addition of one state variable for each objective, but the number of decision variables remains the same.

Implicit stochastic optimization determines the optimum decision sequence for each of many generated streamflow sequences through the application of a deterministic optimization technique. The set of optimum policies is examined by multivariate analysis and a decision function is extracted. This release rule is then used to estimate the optimum decision for the reservoir design or real time operation. The Monte-Carlo DP procedure is identical to that described above with DP being used as the deterministic optimization technique [*Young*, 1967; *Jettmar and Young*, 1975; *Bhaskar and Whitlatch*, 1980].

Two stochastic DP approaches have been used in solving reservoir operation problems: the Markov chain method of successive approximation and the policy iteration algorithm [*Howard*, 1960]. The method of successive approximation is a straightforward generalization of deterministic DP to the nonstationary stochastic case. Howard's algorithm uses the value determination operation and policy improvement routine for a (cyclic) stationary policy in infinite horizon problems. These stochastic DP methods have been applied to the reservoir management and operation by *Butcher* [1968, 1971], *Loucks and Falkson* [1970], *Dudley and Burt* [1973], *Torabi and Mobasheri* [1973], *Su and Deininger* [1974], *Mawer and Thorn* [1974], and *Turgeon* [1980, 1981]. The successive approximation has some advantages over the policy iteration in terms of computation time and ease of implementation.

Sequential stochastic optimization [*Croley*, 1974] is a heuristic methodology. It is similar to stochastic DP in that the probability distribution of inputs is empirically transformed into that for optimum decisions at each stage. However, instead of maximizing the "expected" total return, the total return is maximized for each of several possibilities of future inflow sequences, as in implicit stochastic optimization, to determine the empirical distribution of the optimum decision at each state. It is a quasi-adaptive process for optimization of real time operation, since single-stage decisions are determined as they are needed by generating streamflow sequences for the length of periods shorter than the whole operation horizon. *Maidment and Chow* [1981] treated the random nature of inflows to the system by incorporating a stochastic inflow model directly within the DP procedure. Transition probabilities found from the resulting stochastic DP were employed to determine the steady state probability distribution of the state and decision variables.

All of the stochastic techniques have been used mainly in the optimization of single-facility multipurpose or multifacility single-purpose reservoir systems. The use of stochastic DP for optimization of multireservoir systems is usually accompanied by the assumption that the various natural inflows into the system are not cross-correlated. This assumption is made in order to reduce the dimension of the DP problem. The use of this assumption, however, results in solutions which are only rough estimates of the optimum design or operation policy that would be achieved if cross-correlation of inflows were considered. Cross-correlation of inflows can probably be best handled by a separation of DP optimization and streamflow

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generation or by the aggregation/decomposition method [*Turgeon*, 1980] which is quite similar to the two-at-a-time optimization of multilevel incremental DP.

NONLINEAR PROGRAMMING (NLP) MODELS

Nonlinear programming (NLP) has not enjoyed the popularity that LP and DP have in water resource systems analysis. This is particularly due to the fact that the optimization process is usually slow and takes up large amounts of computer storage and time when compared with other methods. The mathematics involved in the nonlinear models is much more complicated than in the linear case, and NLP unlike DP cannot easily accommodate the stochastic nature of inflows to the system.

NLP does offer, however, a more general mathematical formulation and may provide a foundation for analysis by other methods. NLP can effectively handle a nonseparable objective function and nonlinear constraints which many programming techniques cannot. NLP includes quadratic programming, geometric programming, and separable programming as special cases which can be used iteratively as a master program or as a subprogram in large-scale system problems. Search techniques have also been used in conjunction with simulation in order to evaluate the performance functions of alternative systems [*Maass et al.*, 1962]. For a system of reservoirs, the number of constraints is large because they deal with similar subsystems, repeated in time or location. Therefore NLP will gain its practical importance in water resources systems with the development of computer technology and effective algorithms for large-scale, multiobjective optimization [*Cohon and Marks*, 1975; *Haimes*, 1977].

Consider a reservoir system problem defined as

$$\min_{l \le X \le u} f(X) \tag{41}$$

subject to

$$g(X) = b \tag{42}$$

in which X is a decision vector of release and storage. The equality constraints include the system of continuity equations and the b vector includes reservoir inflows. The vectors l and u of the inequality constraints represent lower and upper bounds on decision variables. For example, the lower bounds might be dead storage for sedimentation and the minimum downstream requirement for water quality. The upper bounds might be the free board requirement for flood control and the downstream channel capacity for erosion control. The objective function might be in terms of long-term water supply and hydropower production purposes combined with short-range operational goals or targets. The reservoir inflows are usually assumed to be deterministic in most NLP approaches and thus it is desirable to include sensitivity capability in any solution algorithm as emphasized by *Rosenthal* [1980].

With this general NLP formulation, penalty and/or barrier solution methods [*Fiacco and McCormick*, 1968] could be one of the choices. The inequality constraint can be dealt with by using an interior point barrier function and the equality constraint can be dealt with using an exterior penalty function. Then, the augmented objective function becomes

$$V(X; r, t) = f(X) + \sum_{j=1}^{n}\left(-\frac{1}{x_j - u_j} - \frac{1}{l_j - x_j}\right) + t\sum_{j=1}^{m}[g_j(x_j)]^2 \tag{43}$$

where r and t are parameters.

If nonlinear constraints are linearized, the nonlinear problem can be further simplified to

$$\min_{l \le X \le u} f(X) \tag{44}$$

subject to

$$AX = b \tag{45}$$

For this NLP problem with linear constraints, one of the primal methods [*Luenberger*, 1973] or their modification can be used. *Rosen*'s [1960] gradient projection method is motivated by a desire to implement the feasible direction algorithm while not requiring the solution of a linear program at each step. Since the set of active constraints changes by at most one constraint at a time, it is possible to calculate a required projection matrix from the previous one by an updating procedure.

Instead of assuming linearity of equality constraints, both the objective function and constraints might be separated into sums of functions of the N individual reservoirs. Then, the general NLP becomes one of separable problems, probably with some coordinating variables [*Haimes*, 1977]:

$$\min_{l \le X \le u} \sum_{i=1}^{N} f(x_i) \tag{46}$$

subject to

$$\sum_{i=1}^{N} g(x_i) \ge 0 \tag{47}$$

This separable (convex) problem is ideally suited to dual methods, because the required unconstrained minimization decomposes into small subproblems.

Literature from the past dozen years or so does not contain much on the use of nonlinear programming techniques in the operation of reservoir systems. *Lee and Waziruddin* [1970] worked with a theoretical system of three reservoirs in series. The objective function was the maximization of a nonlinear function of irrigation release and water storage in the reservoir subject to linear constraints. The first method they used to solve this problem was the gradient projection method but no step-size term was introduced in the paper. The main disadvantage of the gradient technique was its slow rate of convergence at points near a local optimum. On the other hand, the quasi-Newton method (for unconstrained problems) allows a much faster convergence rate because it considers not only the gradient but also the curvature of the objective function. Thus a quasi-Newton method was applied to the constrained water resources problem outlined above by handling constraints through projection. The main difference between the two algorithms is the premultiplication of the vector gradient $GRADf(x)$ with a modified projection matrix P_q

$$P_q = [I - H^{-1}A_q{}^T(A_qH^{-1}A_q{}^T)A_q]H^{-1} \tag{48}$$

where A_q is the current basis matrix, and H is the Hessian matrix. Although the extended method can lead to substantial improvement in convergence rates, one must recognize that the detailed logic for the generalization can become lengthy. Because the formulation was a linear dynamic and quadratic criteria problem with upper/lower bounds, an optimal control-oriented technique might have been used.

Hicks et al. [1974] used a penalty function technique for optimization of operations for the Pacific Northwest hydroelectric system of the Bonneville Power Administration. The objective was to maximize system energy capability while keeping acceptable uniformity in the surplus of power over-

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load for each time interval. This can be stated as

$$\min f(S, R) = D + W \sum_{j=1}^{J} (D_j - D)^2 \tag{49}$$

in which S and R are the reservoir storage and release vectors, $D_j = L_j - P_j$ is the difference between the total system load requirement L_j and the total power produced P_j for time interval j, and $W > 0$ is a suitable weight. The value of D is determined by

$$D = \frac{1}{T} \sum_{j=1}^{J} D_j T_j \tag{50}$$

where T_j is the length of time interval j, J is the total number of time intervals, and $T = \sum_{j=1}^{J} T_j$ is the total time. The constraints on the system are

$$\begin{gathered} AR + BS + C = 0 \\ R^{\min} \le R \le R^{\max} \\ S^{\min} \le S \le S^* \le S^{\max} \\ E_{i,j-1} - E_{i,j} \le T_j \Delta E_i^{\max} \\ P_{i,j} \le 0.85 P_{i,j}^{\text{peak}} \end{gathered} \tag{51}$$

in which A, B are matrices of 1, 0, -1 values, and C is the matrix of the inflows corresponding to the continuity equations. $E_{i,j}$ and $P_{i,j}$ are elevation and power generated at the ith reservoir during the jth period, S^* is the storage required for desired flood control levels, and $\Delta E_i^{\max}$ is a given constant.

The first step is to divide the constraints into "hard" and "soft" constraints, hard constraints being those that due to physical limitations, cannot be violated in any instance. For this problem, the hard constraints are conservation of mass and reservoir storage bounds. The soft constraints are transferred to the objective function as penalty terms. The penalty terms for any bounded variables, say $x \le x^{\max}$, is defined as

$$V = \{\max (x - x^{\max}, 0)\}^2 \tag{52}$$

and the modified objective function is

$$f(S, R) = F(S, R) + \sum_I W_i V_i \tag{53}$$

in which the weights $W_i > 0$, and i range over all soft constraints. The problem was further simplified by eliminating R through manipulation of the conservation equation. All that was left was the modified objective function and the bounds on reservoir storage. The resulting problem was then solved by the conjugate gradient method extended to handle the remaining hard constraints. While the solutions may not have been the global optimum, they did represent improved values over the initial estimates. A sequence of solutions in which the weights W_i progress from small to large values are normally required for a relatively rapid convergence toward an optimum.

Assuming convexity, rather than linearity of the constraints, leads naturally to the Lagrangian dual procedure. *Chu and Yeh* [1978] derived a modified gradient projection technique for an hourly operation model which they applied to the Shasta Reservoir in Northern California. The hourly model for maximization of the weighted power output was

$$\max \sum_{i=1}^{24} W_i P_i = \sum_{i=1}^{24} W_i N a (R_i/N)^b \{f(S_i) - g(R_i)\}^c \tag{54}$$

subject to

$$\begin{gathered} S_{i+1} = S_i + I_i - R_i \\ S_i^{\min} \le S_i \le S_i^{\max} \\ \bar{R}^{\min} \le R_1/N \le \bar{R}_i^{\max} \\ P_i \ge PS_i \\ \sum_{i=1}^{24} R_i \ge DR \end{gathered} \tag{55}$$

in which a, b, c are constants, N is the number of generation units in use, and $f(S_i)$ and $g(R_i)$ are water surface elevation and tailwater elevation, at the start of the ith hour, respectively; I_i is the hourly inflow, PS_i is the deterministic power schedule, and DR is the total plant release determined by the daily model. The conservation constraint can be written in the form used by *Hicks et al.* [1974], allowing the power function to be converted to a function of release only. If this new power function is assumed to be concave, the objective function is convex, and the Lagrangian procedure can be used.

The reduced gradient method was used by Tennessee Valley Authority [*TVA*, 1976] for scheduling weekly releases. *Hanscom et al.* [1980], in their study of a large hydroelectric system, showed that the search direction is a projection of the reduced gradient. *Divi et al.* [1979] have applied the conjugate gradient method to a hydroelectric power system in Canada. *Rosenthal* [1981] used a modification of the reduced gradient obtained through the solution of an integer programming subproblem for maximizing the benefits in a hydroelectric power system.

Recently, *Simonovic and Marino* [1980] applied the gradient projection method with a two-dimensional Fibonacci search to solve a reliability programming problem for single reservoir management. They considered both random inflow and demand in their continuity equation, and both benefit and risk in their objective function for a discrete determination of reliability concerning flood and drought.

Rosenthal [1980] identified the incompatibility of stochastic inflow and nonseparable benefits within existing mathematical models of reservoir systems. He recommends as the best approach a truncated-Newton algorithm for future development of extensive sensitivity analysis of natural inflows, within a nonseparable nonlinear programming model. A breakthrough in computational efficiency will have to be achieved in order to make this approach possible since the present state-of-the-art is not advanced enough to make feasible more than a handful of sensitivity analyses within a reasonably sized system.

SIMULATION MODELS

Simulation is a modeling technique that is used to approximate the behavior of a system on the computer, representing all the characteristics of the system largely by a mathematical or algebriac description [*Ackoff*, 1961; *Maass et al.*, 1962]. It is different from a mathematical programming technique. Mathematical programming techniques find an optimum decision for system operation meeting all system constraints while maximizing or minimizing some objective. On the other hand, the simulation model provides the response of the system for certain inputs, which include decision rules, so that it enables a decision maker to examine the consequences of various scenarios of an existing system or a new system without actually building it. A mathematical programming model usually requires assumptions on model structure and system

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constraints for practical implementation, whereas a simulation model is more flexible and versatile in simulating the response of the system. On the other hand, optimization looks at (implicitly) all possible decision alternatives, while simulation is limited to a finite number of input decision alternatives. A typical simulation model for a water resources system is simply a model that simulates the interval-by-interval operation of the system with specified inflows at all locations during each interval, specified system characteristics and specified operating rules [*Beard*, 1972].

Loucks and Sigvaldason [1982] have summarized reservoir operating rules that have generally been adopted. For reservoirs in series, in order to reduce unnecessary lower reservoir spilling, downstream reservoirs are depleted before using upstream reservoir water to meet downstream demands. For reservoirs in parallel, water is discharged first from reservoirs with relatively larger drainage areas per unit storage volume capacity. This rule will usually result in a reasonable conservation of water. Another procedure that minimizes expected water spill involves drawing in tandem from each reservoir in a manner that equalizes the probability of reservoir filling for each reservoir. This requires, however, monitoring storage volume and estimating future inflows.

Rule curves are often used in actual system operation [*Loucks and Sigvaldason*, 1982]. For multiple-purpose reservoirs, or for single-purpose reservoirs involving recreation or hydropower, operating policies and associated rule curves commonly define the desired storage volumes and discharges at any time of the year as a function of existing storage volumes, the time of the year, demand for water or hydropower, and possibly the expected inflows [*Loucks and Sigvaldason*, 1982]. Loucks and Sigvaldason also suggest that such operating policies may include one or more of the following four general components.

1. Target storage levels or volumes: these operating rules are limited to a prescription of the desired storage volumes or levels in each reservoir. Reservoir operators are expected to maintain these levels as closely as possible while generally trying to satisfy various water needs downstream.
2. Multiple zoning: operating rules are often defined to include not only storage target levels, but also various storage allocation zones, such as conservation zone, flood control zone, spill or surcharge zone, buffer zone and inactive zone.
3. Flow range: this component of an operating policy provides a more descriptive relationship between reservoir levels and channel flows. The reservoir release and/or diversion depends on which subzone or zone the storage volume is in.
4. Conditional rule curves: conditional rules are defined for reservoir releases not only as a function of the existing storage volumes and time of year, but also as a function of the expected natural inflows into the reservoirs for some prescribed time period in the future.

The earliest simulation model associated with a system of reservoirs appearing in the literature seems to be the study performed by the U.S. Army Corps of Engineers in 1953 for an operational study for six reservoirs on the Missouri River [*Hall and Dracup*, 1970]. Similar simulation studies were performed on the Nile Valley Plain [*Emergy and Meek*, 1960] for problems regarding the size of the reservoirs and their operation. *Maass et al.* [1962] demonstrated the application of simulation techniques to evaluate the economic performance of a river basin.

Most of the simulation models assume that some particular historical streamflow is representative of the entire hydrologic time series. Critical period hydrology has been successfully used for design purposes in a reservoir simulation study [*Askew et al.*, 1971]. *Thomas and Fiering* [1962] suggested use of streamflow simulation to enhance the model reliability. The philosophy behind using a synthetically generated streamflow is that if many sequences of flows are used and each is statistically likely to occur and has a length equal to the operation period of interest, it is likely that the entire range of plausible future scenarios will have been explored [*Burges*, 1979]. *Hufschmidt and Fiering* [1966] used simulation in planning and multireservoir, multipurpose Lehigh River system. The study of 35 configurations resulted in three different designs with higher benefits than the existing system. Synthetic streamflow sequences were used as an input to the system. The sequences were determined from a first-order linear Markov chain model.

Examples of recent applications of simulation include the HEC-3 model developed by the *Hydrologic Engineering Center* [1971], the simulation models (SIM I and II) for the Texas Water System [*Evanson and Mosely*, 1970], the simulation of New York's Oswego Basin [*Liu et al.*, 1972], the Upper Wabash studies at Indiana [*Toebes and Chang*, 1972], HEC-5 model [*Hydrologic Engineering Center*, 1979], Arkansas River Basin model [*Coomes*, 1979], TVA model [*Shelton*, 1979], Hydro System Seasonal Simulator developed by the North Pacific Division, the Corps of Engineers [*Jones*, 1979], Lower Colorado River System [*Freeney*, 1979], Duke Power Hydro System [*Sledge*, 1979], and the Acres model [*Sigvaldason*, 1976].

The HEC-3 model (reservoir system analysis for conservation) and HEC-5 (simulation of flood and conservation systems) are considered to be the best documented of the general reservoir simulation models. The purposes of both models are to simulate the response of water resource systems designed to satisfy simultaneously a variety of water-based needs. They are flexible enough to include any arbitrary configuration of reservoirs and channels. The basic HEC-3 model has been extended to other simulation development (i.e., Oswego Basin model, *Liu et al.* [1972]). HEC-3 has been applied to the operational studies on the Arkansas-White-Red River system in the Southern United States [*Frederich and Beard*, 1972] and also to other river basins [*Eichert and Bonner*, 1979]. The HEC-5 model was developed so that it could perform monthly, daily, or multihourly operations of multipurpose reservoir systems similar to the HEC-3 model. A detailed comparison of these two models was made by Eichert and Bonner. Reservoir system expansion for flood control on the Susquehanna River System was studied using the HEC-5 program [*Eichert and Davis*, 1976]. HEC-5 has been used as an operation model to calculate flood damage in the evaluation of alternative operation plans by *Eichert et al.* [1975].

The simulation models developed by the Texas Water Development Board [*Evanson and Mosely*, 1970] utilize a procedure for minimizing the cost of fluid transfer within a capacitated network. The out-of-kilter LP algorithm for the analysis of capacitated networks is used as the basic mathematical tool. It should be noted that optimization has been used in a simulation model. *Beard et al.* [1972] have compared these models with HEC-3 by applying them on a simplified version of the Texas Water System which consists of two major canal systems containing nine reservoirs and a major import facility and three other major delivery points. The principal differences are as follows.

1. SIM models use a network simulation and solution technique, whereas HEC-3 uses a system search routine and cannot simulate the operation of some types of closed loops without special programming.
2. SIM models use a LP algorithm for computing the op-

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timum operation of the entire system for each computation interval, whereas HEC-3 provides releases in accordance with fixed operation rules based on storage and use schedules. In the case of SIM, incompatible requirements are possible, in which case no solution (infeasible solution) results, and a constraint must be changed manually.

3. Operations are controlled in SIM by a system of delivery and storage priorities (reflecting relative values of water), in relation to specified storage balancing levels, whereas HEC-3 specifies the balance of storage desired among the various reservoirs for any amount of total system storage and provides this distribution of storage subject to water availability in excess of demands at the various locations. Both models produced equivalent results for the specified applications.

The multireservoir model for the Oswego Multilake-River system extends the basic ideas of multireservoir zoning proposed in the HEC-3 model. The scheme used in the Oswego simulation model is based on maintaining all reservoirs at the same zonal position. Downstream flows are adjusted in accordance with the zonal position of the upstream reservoirs.

The Acres model [*Sigvaldason*, 1976; and *Sigvaldason et al.*, 1975] includes the combined rule curve-zoning representation which is inherent in the HEC-3 model. However, this representation was extended by including an additional "spill zone." As with the SIM models, the Acres model uses the out-of-kilter optimization routine as a submodel for achieving optimal responses during individual time intervals. The objective function in the Acres model is designed to reflect the chief operator's optimal decision and monitoring process for a particular operating policy. The Acres model has been used as a day-to-day operating tool for defining reservoir releases in the Trent River System in Ontario [*Sigvaldason et al.*, 1975].

With the aid of a simulation model, one can evaluate the consequence of variations in certain model inputs. On the basis of the results obtained, one can modify the system operation characteristics in such a way as to improve the benefit of the system output. In order to select the best decision, a simulation model has been used conjunctively with an optimization model.

Hufschmidt [*Maass et al.*, 1962] relied on various sampling techniques to generate response surface of the simplified river basin system simulation model, which is virtually an attempt to search for an optimum decision. In view of the complex nature of functions relating water resource system inputs to objective value function, *Beard* [1972] considers a search technique as the most practical overall optimization technique in conjunction with a deterministic simulation model and stochastic inputs. *Jacoby and Loucks* [1972] suggested the use of an optimization model to screen the set of possible plans and to select a small number of them worthy of simulation.

Sigvaldason [1976] considered optimizations as a submodel nested in a simulation model. A time-based rule curve was prescribed to represent ideal reservoir operation. Ranges were prescribed for channel flows, which were dependent on water-based needs. The underlying concept of the model was the mathematical representation of the chief operator's perception of optimum operation and the derivation of this solution using a nested optimization submodel. Penalty coefficients were assigned to those variables which represented deviations from ideal conditions. Different operational policies were simulated by altering relative values of these coefficients. The development and use of the model were simplified by representing the entire reservoir system in a "capacitated network" form and deriving optimum solutions for individual time periods with the "out-of-kilter" algorithm.

REAL TIME OPERATION

Real time reservoir operation concerns the optimal operation of an existing reservoir system. A real time optimization model generally is operated on the basis of forecasted information. Forecasts of streamflow and other input deteriorate with time. A consequence is that both strategic and tactical decisions based on the best possible information are necessary, and an overall decision model for real time operation can be considered as decomposable into long- and short-term models. These models require corresponding reservoir inflow forecasts, and the long-term forecast may be necessarily based on historical patterns but is nevertheless deterministic. Commonly, the long-term model looks ahead over a time space of a year with seasonal, monthly, or weekly time increments. It should be updated each time increment. The short-term model looks ahead in accordance with the above time increments (over a time span of 7 to 30 days) with daily or even hourly increments when hydropower is a product, and should also be updated. Of course, the various long- and short-term models go to different levels of detail. The monthly model need not consider regulating reservoirs or time lags in water transit. The daily model must consider these but need not consider individual units in a facility nor the hydroelectric load demand curve for the day. In this case, an hourly model (over 24 hours) will accept the daily release determinations for each facility without further analysis, but will determine the hour-by-hour releases for each individual unit in each plant.

The objective function to be optimized will depend on the dominant purposes for the particular reservoir system. Usually, however, most purposes and objectives will have been framed within the context of long-term contracts, institutional agreements, and even legislation that are difficult to change in the short-term; and, in general, readily provable benefits for exceeding performance corresponding to contractual obligations are confined to the production of electrical power. An excess of such power will normally bring a return by permitting substitution for more costly modes of generation. Consequently, the objective function will often involve a maximization of hydropower generation while still achieving a satisfactory end-of-period storage vector. A roughly equivalent form is the satisfaction of all system constraints with a minimum expenditure of the potential energy of the stored water in the reservoir.

Another type of benefit may be available when the system contains pumped storage plants. These plants have the characteristic of enabling peak hour power capacity to be increased, thus possibly saving the cost of adding another capacity for this purpose. However, it does this at a price; overall efficiency is not more than about 70% and energy is always lost in its operation. To be economical, these pumped storage plants must be applied by energy during off-peak hours from base power plants (such as nuclear or run-of-the-river hydroelectric plants) which are most efficient when supplying their base load 24 hours a day.

The objective function and power and energy constraints arising from various commitments are nonlinear with respect to reservoir release or reservoir end-of-period storage. However, in many situations these nonlinearities can be effectively linearized, and linear techniques, which have the capability of handling large numbers of variables and constraints, can be employed.

Whatever the form of the decision model and the technique used, only approximately optimal policies can be obtained, since almost always, approximations of one sort or another must be made to facilitate practical solutions. The differences

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between various solution techniques are in speed of convergence, computational requirements, the need for an initial feasible policy, and convenience of application. Of course, these differences can be significant. A common problem relates to nonlinearities in constraints and/or objective function and the high dimensionality typical of the reservoir system optimization problem. The latter leads in some applications to a requirement for further problem decomposition.

A mathematical model of a hydropower system can be formulated so as to minimize the effects of nonlinearities arising from power and energy relationships. These nonlinearities are basically a result of the variable head at the turbines consequent to storage reservoir release. For a given release of water through the turbines over some time interval, the average head is the difference between average forebay elevation and tailwater elevation. The former is a function of the average storage over the interval which depends on the release; the latter is a function of the rate of release. In addition, power for a given head is not a linear function of release.

Time decomposition of the optimization procedure, both by long- and short-term models and internally within each model, can materially assist in a linearization process. The effect of variable head is least for the shorter-term models; and the greater detail and the more accurate power and energy calculations can be reserved for hourly optimization in which these calculations are most important, the hourly load curve representing the most significant constraint.

California Central Valley Project (CVP)

The concept of real time optimization of reservoir operation as described in the previous section has been applied to the CVP [*Yeh*, 1979] which consists of 9 reservoirs, 9 power plants, 3 canals, and 4 pumping plants. The overall procedure is to optimize in turn a monthly model over a period of one year, a daily model over a period of up to one month, and an hourly model for 24 hours. Outputs from one model (monthly ending storages or daily releases) are used as inputs into the next echelon model, iterating and updating whenever new information on streamflow prediction becomes available.

The hourly model is the one that is used for scheduling hourly reservoir releases. It maximizes a weighted summation of generated power over a 24-hour period subject to specified plant releases obtained from a daily model, a desired hourly power schedule, and appropriate system and equipment constraints. The optimization procedure involves an iterated LP allied with IDPSA. CVP operational records of some typical days were compared with the hourly model outputs for those days, and significant improvements were observed, both in better conformance with the desired power schedule and more efficient hydropower production.

Tennessee Valley Authority (TVA)

In 1974, TVA started a project to develop a comprehensive water resource management method to cope with the increasing complexity of multipurpose reservoir system management. Complete implementation of the project is planned for 1985. An overview of the project plan is given by *Shelton* [1979] and *Wunderlich* [1979]. The major purpose of the TVA system is to provide navigation and flood control, and so far as consistent with these two, power generation from 42 TVA-operated reservoirs. The project is subdivided into four major segments: weekly planning, weekly scheduling, daily planning and daily scheduling. The planning models are used for basic studies, off-line in a batch mode, while the scheduling models address the day-to-day operational problems.

The weekly planning model, a DP with weekly time steps over planning periods of up to one year, analyzes selected system configurations for specified inputs and demand and evaluates five operation objectives and calculates storages in 19 reservoirs and releases from 42 reservoirs so as to minimize a system performance index. The weekly scheduling model uses DP to calculate long-range operation guides by weekly time steps for planning periods of up to one year, uses LP for week-by-week calculation of reservoir levels and releases, and considers all active operating constraints and guides by specified priority. The daily planning model analyzes a selected system configuration by daily (or hourly) time steps, simulates present policies or finds optimal policies by optimizing a specified criterion, and uses varying levels of detail of system simulation. The daily scheduling model aids in scheduling reservoir levels and releases by daily (or hourly) time steps over weekly or biweekly periods in conjunction with weekly models, and consists of a set of weekly, daily and hourly scheduling models and supporting modes (as required for day-to-day scheduling) for interactive use. The TVA model includes simulation of the physical system, evaluation of operating purposes, real time operation, forecasting models for system inputs and demands, and input and output data management.

Shane and Gilbert [1982] and *Gilbert and Shane* [1982] have presented the results of a major effort to develop computer program and mathematical methods needed to model the week-to-week variations in important scheduling characteristics of the TVA system. The system of preemptive priority operating constraints was implemented using LP with upper bounding to save computer time and storage. The nonlinear power cost function then was minimized subject to the system of priority constraints. A guide development program was formulated by a standard stochastic DP that minimizes the expected future power cost.

Hydro Quebec

A linear separable programming model was developed to integrate the operation of the Ottawa River basin [*Bechard et al.*, 1981]. The purposes of the model are to control floods and to generate hydroelectric energy. Due to constraints imposed by conflicting purposes, trade-offs between these two purposes are analyzed by multiple-objective optimization. The basic approach is based on a composite objective function incorporating weighting coefficients on the two objectives. Parametric variation of the weights leads to an approximation of the set of noninferior (nondominated, Pareto-optimal) solutions.

Due to the different environment required for attaining each objective, a hierarchical approach is needed, i.e., the long-term model, the midterm model, and the short-term model. All three models are to be run sequentially with updated runoff forecasts. The entire process is sequentially forwarded in time. The updating intervals are not fixed and left at the user's discretion. All three models use a common computer program except the differences in the length of the time horizon, the time periods, the target storages, and the objective function weights. Each linear-separable program is developed and is solved by an IBM software optimization package. As a result of the computational experience, real time operation of the system has been recommended and is in process.

Turgeon [1981] reviewed some of the recently developed methods for optimal short-term scheduling of hydroplants in series which include the progressive optimality algorithm, the discrete maximum principle and Rosen's gradient projection method. On the basis of his analysis, it appears that the pro-

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gressive algorithm is best suited for the short-range scheduling problem.

CONCLUSIONS AND RECOMMENDATIONS

LP Models

LP has been a valuable tool in the optimization of reservoir operations, both for real time and planning purposes. Although objective functions as well as some of the constraints are often nonlinear, various linearization techniques, such as piecewise linearization, first-order Taylor series expansion and iterative schemes, can be successfully used.

The essential advantages of LP include (1) its ability to accommodate relatively high dimensionality with comparative ease, (2) universal optima are obtained, (3) no initial policy is needed, and (4) standard computer codes are readily available. Decomposition techniques are necessary for large, complex reservoir systems where the numbers of operating variables and constraints are high. Combinations of LP and DP have also been used to alleviate the problem of dimensionality. In some special cases, reservoir operation problems have been solved successfully with out-of-kilter and other network algorithms.

Suggested future research includes (1) developing and implementing decomposition schemes, such as the Dantzig-Wolfe decomposition, for practical application; (2) resolving the dimensionality problem associated with the stochastic LP approach; and (3) continuing the development of stochastic programming with recourse in the application to reservoir operation.

DP Models

DP is capable of handling adaptive, nonlinear, and stochastic problems in reservoir systems, DP is specifically applicable to those reservoir planning and operation problems which can be represented as either a progressive or serial directed network problem. Among the assumptions of monotonicity and separability in the DP model, the separability condition of the objective function limits some model applications and thus an artful choice of stage and state is required. In addition to the multipurpose and multifacility nature of a large reservoir system, considerations of reliability and real time operation have aggravated the so-called curse of dimensionality, and thus numerous efforts have been attempted to alleviate this problem during the past decade.

In deterministic operation, two lines of development can be observed which is based on either IDP or DDP. IDP has been extended to IDP with successive approximation (IDPSA) and then to multilevel IDP; DDP has been extended to discrete DDP (DDDP) and then to constrained DDP. DP-LP, multiobjective DP, and progressive optimality algorithm are other lines of approach to reservoir release scheduling or planning problems. Both constrained DDP and progressive optimality algorithm have a rather unique feature in that the state space does not have to be discretized.

In stochastic operation, stochastic DP and sequential stochastic optimization are applicable for stationary policy and real time operation, respectively. Monte Carlo DP and chance-constrained DP have the important advantage of being able to assess the reliability or risk associated with a design or planning policy. The application of these probabilitistic formulations to a multipurpose multifacility reservoir system is either time consuming or difficult to achieve. The techniques used in handling the dimensionality problem in deterministic DP might be applied to reduce these problems.

The following five areas of future research are recommended: (1) continuation of research in IDP procedures to include risk or reliability and multiobjectives; (2) development of research in DDP procedures to facilitate nondifferentiable objective functions; (3) continuation of research in real time operations to incorporate adaptive forecasts in an interactive mode; (4) development of research in decomposition or partitioning to facilitate nonseparable objective functions; and (5) initiation of research in semi-Markov programming for multiyear reservoir operation with an econometric forecast.

NLP Models

Basically, there are two methods that have been used for reservoir systems: the conjugate gradient-gradient projection method extended to handle linear constraints with penalty terms for nonlinear constraints, and the Lagrangian gradient procedure for convex programs. The few articles on the subject confirm that the programs work, but rate of convergence and computer requirements for major obstacles. Overall, therefore it would appear that NLP techniques, while effective, are not really viable at the present time unless the separability condition is assumed; and, of course, bigger and faster computers are being built, so that the dimensionality problem we face now will become less severe in the future.

Areas of interest for future research are (1) development of some accelerated procedures for existing methods for faster convergence; (2) incorporation of streamflow sensitivity features within both primal and dual algorithms; (3) extension of reliability programming to the multiple-reservoir problem; (4) development of effective algorithms for both spatial and temporal decomposition without destroying nonseparability; (5) application and comparison of the cutting plane and the tangential approximation procedures with existing methods; and (6) incorporation of dynamic supply-demand relationships with changes of price in the objective function.

Simulation Models

Simulation models have been successfully used by various practitioners. However, in recent years a tendency has been toward incorporating an optimization scheme into a simulation model to perform certain degrees of optimization. As a result, the distinction often made between the optimization and simulation tends to be obscured. It has been quite common to have a few optimization routines nested in a simulation model. *Eichert* [1979] pointed out that from the practitioner's point of view, mathematical programming techniques have, thus far, not proven to be widely useful because of the complexities of water resources systems and noncommensurable objectives in water resources management. In this regard, simulation is an effective tool for studying the operation of the complex water resource system incorporating the experience and judgment of the planner or design engineer into the model. It would be desirable for the simulation model to have some capability of selfoptimization like the model of *Sigvaldason* [1976] to reduce the amount of computation to obtain an optimum or near optimum operation plan for a complex reservoir system.

Real Time Operation

One of the important conclusions made at the ASCE National Workshop on Reservoir Systems Operations (1979) was that although considerable resources were being expended in systems analysis research the resultant techniques, tools, and data were being tested, adapted, and applied very slowly. Although some successful applications of systems analysis tech-

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niques have been reported in the literature, the reservoir operators are still very reluctant to use optimization models for their day to day scheduling of water releases and power generations. The reasons can be summarized as follows.

1. Most of the reservoir operators have not been directly involved in the development of the computer model and thus are not entirely comfortable in using the model, particularly under the situation where modifications have to be made in the model to respond to changes encountered in the day to day operation,

2. Most of the published papers deal with simplified reservoir systems and are difficult to adapt for use in a real system. In addition, most of the published research results are poorly documented from the practical use point of view,

3. There are institutional constraints that impede user-research interactions.

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